Data Audit Report: Predicting Engine Life

Analytics Capstone Project (ANCP) | Deliverable D2 | Module M2: Data Acquisition and Preparation | Group P | TCS (analytics partner) for GE Aerospace (client) | Data version **data-v1.0** | Lead: Ashish Kumar Shrivastava (BA05010); reviewers: Bittu Kumar (BA05013), Boomikaa P (BA05014)

0. Sprint 2 summary (Agile / Scrum view)

Product goal (from D1): decide *when to remove each engine* using three analytics layers: **predictive** (RUL forecast), **prescriptive** (repair-timing rule that picks the removal threshold) and **simulation** (Monte Carlo cost comparison against a fixed-interval schedule). **Sprint 2 goal**: deliver trustworthy, leakage-safe, reproducible data that all three layers can use. Sprint ran 16 to 23 Oct 2026 (charter target 21 Oct, portal deadline 23 Oct). Roles: Product Owner = course faculty/client proxy; sprint lead = Ashish; peer review = Bittu and Boomikaa.

ID	User story (as an analyst/decision-maker I need...)	Acceptance criteria (Definition of Done)	Pts	Status
S2-1	provenance, licence and a DPDP statement so the data may be used	Sec. 1 to 2 filled; citation and access date recorded	2	Done
S2-2	a reproducible quality audit	Script rebuilds data-v1.0 from raw; missing, duplicate, gap and outlier checks logged	3	Done
S2-3	representativeness and bias checks	Train vs test, regime and lifetime coverage tested; gaps stated	2	Done
S2-4	a feature log with leakage review	Every new column has logic and leakage risk; second member signs off	2	Built; sign-off pending
S2-5	datasheet, data dictionary, checksums	Appendix A, data_dictionary.csv, SHA256SUMS.txt in repo	2	Done
S2-6	simulation- and prescriptive-ready inputs	Lifetime distributions fitted (Sec. 6) for the D6 simulator	1	Done

Sprint review (what the audit found). (1) Data is complete: 0 missing values, 0 duplicates, 0 cycle gaps in all 8 files. (2) The charter has the FD004 engine counts reversed: the files hold **249 training and 248 test** engines, not 248 and 249 (corrected here; fix in D1 v2). (3) FD001 has 7 constant columns that carry no information (dropped); FD004 has none. (4) Flagged outliers are mostly late-life degradation, so they are **kept, not deleted**. (5) Train and test lifetimes are statistically indistinguishable (KS p = 0.99 and 0.86), but everything is simulated, one engine type, no real maintenance events.

1. Source and access

Item	FD001	FD004
Dataset	NASA C-MAPSS Turbofan Engine Degradation Simulation (Saxena and Goebel, 2008), NASA Prognostics Center of Excellence, Ames Research Center	same release
Files used	train_FD001, test_FD001, RUL_FD001 (.txt)	train_FD004, test_FD004, RUL_FD004 (.txt)
Location	data.nasa.gov/dataset/cmapss-jet-engine-simulated-data (official). Working copy fetched 23 Oct 2026 through the public GitHub mirror github.com/ericlrf/rul; row and engine counts match the NASA readme	same
Licence / permission	Free public NASA open data for research. The download carries no separate licence file; the team treats it as public U.S. government data and cites it as above. No permission request needed	same
Date accessed	23 Oct 2026	23 Oct 2026
Integrity	Raw files stored unmodified in raw/; SHA-256 of each file in SHA256SUMS.txt (verify against a fresh NASA download)	same

Citation: A. Saxena and K. Goebel (2008). "Turbofan Engine Degradation Simulation Data Set", NASA Ames Prognostics Data Repository. Also A. Saxena et al. (2008), "Damage propagation modeling for aircraft engine run-to-failure simulation", PHM08.

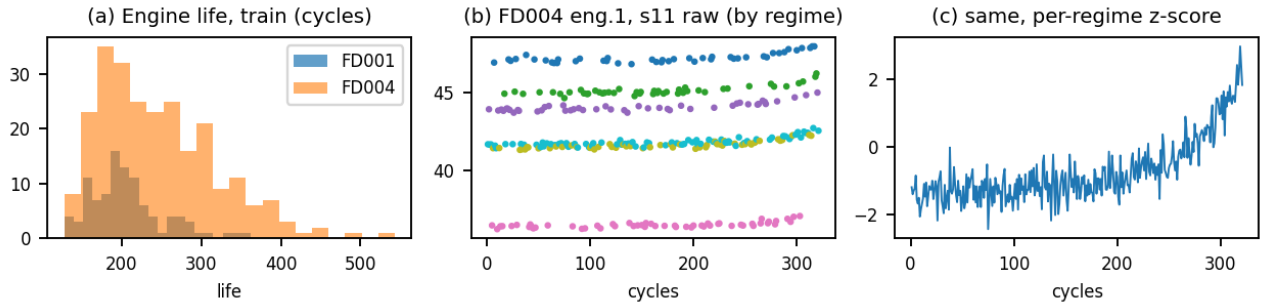
2. Consent and purpose limitation (DPDP Act, 2023)

The DPDP Act, 2023 governs *digital personal data*, that is, data about an identifiable individual. The C-MAPSS files hold only engine ID, cycle count, three operating settings and 21 simulated sensor values per cycle. There is no name, crew, passenger, flight, tail-number, airline or location field, and the values are produced by a simulator, not collected from people. **Conclusion:** the dataset is outside the Act, so no consent or notice is required. **Purpose limitation:** the data is used only for this academic study of RUL prediction, repair-timing rules and cost simulation, and will not be linked to any other source. **If GE fleet data is used later**, a fresh review is needed: tail numbers, flight routes, crew-linked logs or operator identities could make records personal data, and contractual confidentiality would also apply. (Team statement, not legal advice.)

3. Quality audit

Check	FD001 train / test	FD004 train / test	Treatment and reason
Rows (engines)	20,631 (100) / 13,096 (100)	61,249 (249) / 41,214 (248)	None. Counts verified against NASA readme
Missing values	0 / 0	0 / 0	No imputation needed; script raises a warning if this changes
Duplicate rows; duplicate (engine, cycle)	0; 0 / 0; 0	0; 0 / 0; 0	None needed
Cycle gaps (must run 1..n)	0 / 0	0 / 0	None needed; rolling windows in D4 are therefore safe
Wrong data types	0	0	All 26 columns numeric
Constant (zero-variance) columns	7: op3, s1, s5, s10, s16, s18, s19	none	**Dropped** from cleaned file (FD001): no information, and they break z-scoring
Near-constant (<=5 distinct values)	s6	op3, s16, s19	Kept raw; excluded from the z-score outlier rule, which is invalid for discretised values
Outlier rows, |z|>4 within regime (train)	241 (1.2%) / 2	1,174 (1.9%) / 493	**Retained and flagged** (outlier_flag); see below

Why outliers are kept. Gaussian noise would give about 0.006% of readings beyond |z|=4, but we see more, concentrated in s9 and s14 (FD001) and s7 to s14 (FD004). These rows sit at the *end of life*: in FD001 all flagged rows have RUL <= 50 (median RUL 6, versus 103 overall) and in FD004 67% do (median 11, versus 122), against only 25% and 21% of all rows. They are the degradation signal the model must learn, so removing them would delete exactly the cases that matter for failure risk. Method: z-scores per operating regime, so that FD004's six regimes are not mistaken for outliers (panels b and c below). Flag only; no clipping or deletion.



4. Representativeness and bias checks

The data describes machines, not people, so "bias" here means **coverage bias**: which engines, conditions and failures are missing or thin.

Check	Result	What is under-represented / implication
Train vs test lifetime	FD001 life: train mean 206.3, test mean 206.5 (KS p=0.99). FD004: 246.0 vs 252.7 (p=0.86)	No distribution shift between our train and test splits, so honest test error is credible. This says nothing about real GE engines
Long-life engines	FD001 life: median 199, max 362. FD004: median 234, max 543 (sd 73)	Right-skewed: very long-lived engines are rare, so RUL may be under-predicted for them. Report error by life band in D5
Operating regimes (FD004)	Train share per regime: 15.1%, 15.1%, 14.8%, 14.9%, 15.0%, 25.1%; test almost identical; 0 unseen test regimes	Regime 6 holds 25% of rows, others about 15%: mild imbalance. Regime-wise errors to be reported. FD001 has a single regime only
Failure modes	FD001: one (HPC degradation). FD004: two (HPC and fan). Sensor-fault, bird-strike or maintenance events are absent	Real fleets mix more failure types, ageing, repairs and swaps; proof of concept only (charter risk 1)
Early-life dominance	39% (FD001) and 49% (FD004) of training rows have RUL > 125	Labels are skewed to healthy engines; hence RUL_cap = 125 is provided and metrics must be checked near end of life
Truncated test histories	Test engines observed for 31 to 303 cycles (FD001) and 19 to 486 (FD004); final RUL 7 to 145 / 6 to 195	Some test engines have very short history (19 cycles): windows must be padded, and short-history performance reported separately
Other gaps	Single simulated engine class; no airline, climate or route diversity; sensor noise is synthetic	Do not claim fleet-wide validity; re-validate on GE data (future phase)

5. Feature engineering log

Column	Logic	Leakage risk and control
RUL (target)	Train: last cycle minus current cycle. Test: true final RUL (RUL_FDxxx.txt) plus cycles remaining to the last observed one	Uses the future by definition. **Target only, never an input.** Test RUL is for scoring only
RUL_cap	min(RUL, 125), a common convention that reflects negligible degradation early on	Target only. Cap value is a D4 tuning choice, not fitted on test
condition_id	Regime from rounded op1 and op2 (FD001: 1; FD004: 6)	Low. Mapping defined on train; all test rows matched (0 unseen)
s*_z (per sensor)	(reading minus regime mean) / regime sd, statistics from the **train file only**	Statistics never use test rows. In D5 CV, recompute them inside each training fold
outlier_flag	1 if any |z| > 4 in the row	Row-level, no future information. Predictive use is optional
Dropped columns	FD001 constants (7), see Sec. 3	None
Planned for D4 (not in data-v1.0)	Rolling mean/slope over past k cycles; cycle count as input	Windows must look backwards only. **Forbidden:** cycle divided by total life, row-wise train/test splits. Engine key is (subset, split, unit): unit IDs repeat across files

6. Readiness for predictive, prescriptive and simulation work

Layer	Question	Data input from D2	Sprint
Predictive	How many cycles remain for each engine?	Cleaned z-sensors, RUL and RUL_cap; baseline = mean life minus current cycle (FD001 mean life 206.3; FD004 246.0)	S3 to S4 (D3, D4, D5)
Prescriptive	At what predicted RUL (threshold tau) should we remove an engine to minimise expected cost?	Prediction error from D5; decision variable tau; constraint: failure probability cap; cost ratio failure:early removal = 10:1 (charter), swept 3 to 30	S5 (D6)
Simulation	What does a fleet cost under the rule versus a fixed interval?	Monte Carlo draws of engine life from Weibull fits to train lifetimes: FD001 shape 4.41, scale 225; FD004 shape 3.47, scale 273 (cycles), plus model error noise; 1,000+ runs per setting	S5 to S6 (D6, D7)

Shape above 1 means wear-out failures, which is what makes timed removal worthwhile; a fixed interval must be set conservatively below the weakest engines (minimum life 128 cycles), while a RUL-based rule can follow each engine. Exact cost formulas are fixed in D6 and every assumption will be stated.

7. Retrospective (draft; edit after the team meeting) and next sprint

Went well: one script rebuilds everything from raw data, so the audit is repeatable. **To improve:** verify the data against a fresh NASA download, and log the meeting dates in D1 Appendix A. **Actions:** Ashish tags the repository data-v1.0 and confirms checksums; Bittu signs off the leakage review (S2-4); Boomikaa pulls the s11/s4 trends (strongest correlation with capped RUL, |r| about 0.7 to 0.8 in both subsets) into the D3 EDA notebook, due 28 Oct (target 26 Oct). Next sprint backlog: S3-1 EDA of sensor trends by regime; S3-2 correct charter engine counts (D1 v2); S3-3 baseline and first candidate models.

Appendix A. Datasheet for datasets: C-MAPSS FD001 and FD004, data-v1.0

Following the "Datasheets for Datasets" structure (Gebru et al.). Completed by Group P on 23 Oct 2026.

Motivation	
Purpose	Created by NASA to benchmark prognostics (RUL estimation) for run-to-failure turbofan engines. Our use: predict RUL, set a repair-timing rule and simulate maintenance cost for the GE Aerospace / TCS scenario.
Creator and funding	NASA Ames Prognostics Center of Excellence (A. Saxena, K. Goebel et al.), public research data. Our derived version: ANCP Group P, academic coursework; no funding.
Composition	
Instances	Each row is one flight cycle of one simulated engine. FD001: 20,631 train rows (100 engines), 13,096 test rows (100). FD004: 61,249 train rows (249), 41,214 test rows (248). Train engines run to failure; test engines stop earlier, with final RUL given separately.
Fields	Engine ID, cycle, 3 operating settings, 21 sensors (raw); cleaned files add condition_id, z-scored sensors, outlier_flag, RUL, RUL_cap. FD001 cleaned: 38 columns; FD004: 51. Full list in data_dictionary.csv.
Subsets / conditions	FD001: 1 operating condition, 1 fault mode (HPC degradation). FD004: 6 conditions, 2 fault modes (HPC and fan).
Missing data, noise	No missing values or duplicates (Sec. 3). Sensor noise is synthetic. Sensor units are not published in the files.
Sensitive data	None: no personal, operator or confidential information (Sec. 2).
Known gaps	Simulated engines, single engine class, limited failure modes, no maintenance or sensor-fault events (Sec. 4).
Collection process	
How generated	Produced with NASA C-MAPSS, a high-fidelity simulation of a large commercial turbofan. Each engine starts with different initial wear and manufacturing variation, degrades until a health limit is reached, and sensor noise is added. No people were observed; no consent involved.
Time frame	Released by NASA in 2008. Accessed by us on 23 Oct 2026.
Preprocessing, cleaning, labelling	
Steps	(1) Read raw whitespace-delimited files, name 26 columns. (2) Audit missing, duplicate, gap, type, constant and outlier checks. (3) Drop zero-variance columns. (4) Derive condition_id. (5) Create RUL and RUL_cap labels. (6) Z-score per regime using train statistics. (7) Flag |z|>4 rows (kept). (8) Round to 4 decimals and write CSV.
Raw data kept?	Yes, unchanged in raw/ with checksums. Everything above is rebuilt by clean_data.py (python clean_data.py raw data-v1.0; needs pandas, numpy, scipy).
Uses	
Intended uses	Teaching and proof-of-concept RUL prediction, repair-timing optimisation and cost simulation.
Not suitable for	Real airworthiness or maintenance decisions; claims about GE engines without re-validation on real fleet data; any split by row (always split by engine).
Cautions	Do not use RUL columns as model inputs; test RUL only for scoring; recompute z-statistics inside training folds when cross-validating.
Distribution	
Where, licence	Raw: NASA open data portal (public research use). Cleaned data-v1.0: team repository (private to the course), with README, data dictionary and SHA256SUMS.txt; must keep the NASA citation.
Integrity (SHA-256, first 16)	FD001_test: 0b0a57707fd88bda; FD001_train: a660084356717d53; FD004_test: d878c241eca8af68; FD004_train: 8d7c437c518e4075. Raw file hashes are in SHA256SUMS.txt.

Maintenance	
Owner / contact	Ashish Kumar Shrivastava (BA05010), with Bittu Kumar (BA05013) and Boomikaa P (BA05014), ANCP Group P.
Versioning	Semantic tag data-v1.0. Any change to cleaning rules or columns raises the version (v1.1 minor; v2.0 if columns or label definitions change) and is logged in the repository changelog. Next possible change: RUL cap tuning and rolling features after D4.
Retention / errata	Charter correction: FD004 has 249 train and 248 test engines. Mirror copy to be re-checked against the NASA download.

A.2 Sensor and setting dictionary (NASA C-MAPSS readme)

Col	Symbol	Description	FD001	FD004
op1 / op2 / op3	-	Altitude, Mach number, throttle resolver angle (operating settings)	op3 dropped (constant)	6 regimes via op1, op2
s1	T2	Total temperature at fan inlet	dropped	kept
s2	T24	Total temperature at LPC outlet	kept	kept
s3	T30	Total temperature at HPC outlet	kept	kept
s4	T50	Total temperature at LPT outlet	kept	kept
s5	P2	Pressure at fan inlet	dropped	kept
s6	P15	Total pressure in bypass duct	kept (near-constant)	kept
s7	P30	Total pressure at HPC outlet	kept	kept
s8	Nf	Physical fan speed	kept	kept
s9	Nc	Physical core speed	kept	kept
s10	epr	Engine pressure ratio (P50/P2)	dropped	kept
s11	Ps30	Static pressure at HPC outlet	kept	kept
s12	phi	Ratio of fuel flow to Ps30	kept	kept
s13	NRf	Corrected fan speed	kept	kept
s14	NRc	Corrected core speed	kept	kept
s15	BPR	Bypass ratio	kept	kept
s16	farB	Burner fuel-air ratio	dropped	kept (near-constant)
s17	htBleed	Bleed enthalpy	kept	kept
s18	Nf_dmd	Demanded fan speed	dropped	kept
s19	PCNfR_dmd	Demanded corrected fan speed	dropped	kept (near-constant)
s20	W31	HPT coolant bleed	kept	kept
s21	W32	LPT coolant bleed	kept	kept

LPC/HPC = low/high-pressure compressor; LPT/HPT = low/high-pressure turbine. Units are not published in the data files.

A.3 Cleaned file inventory (data-v1.0)

File	Rows	Columns	Size (MB)	SHA-256 (first 16)
FD001_train_clean.csv	20,631	38	5.1	a660084356717d53
FD001_test_clean.csv	13,096	38	3.3	0b0a57707fd88bda
FD004_train_clean.csv	61,249	51	19.3	8d7c437c518e4075
FD004_test_clean.csv	41,214	51	13.0	d878c241eca8af68
data_dictionary.csv	-	4	<0.01	-
audit_results.json	-	-	<0.01	-

A.4 Summary statistics for simulation inputs

Statistic (cycles)	FD001	FD004
Training engine life: min / median / mean / max	128 / 199 / 206.3 / 362	128 / 234 / 246.0 / 543
Training life standard deviation	46.3	73.1
Test engine total life (observed + final RUL): min / median / max	141 / 199 / 341	126 / 235 / 554
Weibull fit to training life (shape / scale)	4.41 / 225	3.47 / 273
Share of training rows with RUL > 125	38.9%	48.8%